

Homework 2

MEGN544/ROBO554 Robot Mechanics: Kinematics, Dynamics, and Control

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Problem 1. Given the Euler angle set {Roll (ψ), Pitch (θ), Yaw (ϕ)}: RotZ(ϕ)RotY(θ)RotX(ψ):

- (a) Derive the resulting rotation matrix in terms of ϕ , θ , and ψ .
- (b) Provide the inverse solution. In other words, given a rotation matrix,

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix},$$

determine the formulas for ϕ , θ , and ψ in terms of the entries in $\mathbf{R}$.

- (c) At what value of which angle ϕ , θ , and ψ does the inverse solution degenerate? What happens to the remaining two angles?
- (d) Using a sketch of the intermediate rotations to assist, describe why this is a degeneracy.

Problem 2. Given the following rotation matrix,

$$\mathbf{R} = \begin{bmatrix} 0.6533 & 0.6403 & 0.4040 \\ -0.3488 & 0.7282 & -0.5900 \\ -0.6719 & 0.2445 & 0.6991 \end{bmatrix}$$

- (a) Prove that it is a rotation matrix.
- (b) What are the ZYZ angles for this rotation matrix?
- (c) What is the angle and axis representation for this rotation matrix?
- (d) What is the Quaternion representation?

Problem 3. What is the rotation matrix and angle-axis vector ($\mathbf{\Omega} = \hat{\mathbf{k}}\theta$) that are described by the following quaternions,

- (a) $[1, [0 \ 0 \ 0]^\top]$.
- (b) $[0, [0 \ 0 \ 1]^\top]$.
- (c) $[0, [1 \ 0 \ 0]^\top]$.
- (d) $[0.5332, [0.5928 \ 0.08311 \ 0.5978]^\top]$.

Problem 4. An object is spinning with some constant angular velocity ${}^0\boldsymbol{\omega}$ such that after some time t it has rotated $\theta = \|\boldsymbol{\omega}\|_2 t$ radians about the ${}^0\hat{\mathbf{k}} = {}^0\boldsymbol{\omega}/\|\boldsymbol{\omega}\|_2$ direction. At this point, its new orientation is given by ${}^0\mathbf{R}_{obj} = {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj}$. Using angle-axis representation, derive the time derivative of a rotation matrix given an arbitrary angular velocity. In other words, compute $\dot{\mathbf{R}} = d\mathbf{R}/dt$ which by the chain rule yields,

$$\frac{d\mathbf{R}}{dt} = \left(\frac{d}{d\theta} {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj} \right) \dot{\theta} + \left(\frac{d}{d\hat{\mathbf{k}}} {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj} \right) \dot{\hat{\mathbf{k}}},$$

and evaluate at $\theta = t = 0$ before the orientation is modified by rotation.

Hint: Use (3.41) in Hollerbach's notes for ${}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}})$ (Rodrigues' formula) and consider $\dot{\hat{\mathbf{k}}}$ for a constant angular velocity. In addition, evaluate what is $\dot{\mathbf{R}} = \lim_{t \rightarrow 0} \frac{d}{dt} (\mathbf{R})$.

Problem 5. Consider the combination of robot, square table, block, and camera below, with associated coordinate systems as shown. The relative locations of robot, table, block, and camera are shown in Fig. 1. The block is located at the center of the table.

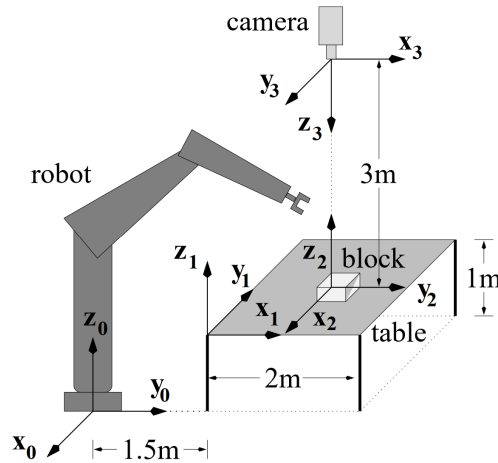


Figure 1

- (a) Find ${}^0\mathbf{T}_1$, ${}^1\mathbf{T}_2$, ${}^2\mathbf{T}_3$, and ${}^0\mathbf{T}_3$ by inspection.
- (b) Assume the block has moved. It is tracked by the camera and is known to exist at location ${}^3\mathbf{d}_{32} = [-0.5 \ 0.5 \ 3]^\top$. Where is the block in the robot's coordinate system, i.e., what is ${}^0\mathbf{d}_{02}$?